



# Modelling Financial Markets

## Seminar 9: Realised Volatility — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, [antoaneta.amza@csie.ase.ro](mailto:antoaneta.amza@csie.ase.ro)

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BV: Bipower Variation
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ DM: Diebold–Mariano (test)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FDR: False Discovery Rate
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-CJ: HAR model with Continuous and Jump components
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ IQ: Integrated Quarticity
- ▣ JB: Jarque–Bera (test)
- ▣ MCS: Model Confidence Set
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ NW: Newey–West (standard errors)
- ▣ OLS: Ordinary Least Squares
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QQ: Quantile–Quantile (plot)
- ▣ RF: Random Forest

## Acronyms used in this material (2/2)

- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SE: Standard Error
- ▣ SHAR: Semivariance HAR model
- ▣ TQ: Tripower Quarticity
- ▣ TSRV: Two-Scale Realised Variance
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index

## Today's Questions and Route

- ▣ Questions of the seminar
  - ▶ **how precisely does one day of intraday prices measure volatility?**
  - ▶ **does the measurement improve a forecast?**
- ▣ Route
  - ▶ Primer, what you need today from [Chapter 9](#) (the lecture comes after this seminar): RV, noise, jumps, HAR, forecast evaluation
  - ▶ Part I, measurement and diagnosis: A1 precision of RV, A3 noise and sampling, B3 sampling grids on SPY, B1 jump days in SPY, A2 one jump in six returns
  - ▶ Part II, models and forecasts: A5 HAR as a restricted AR(22), B5 log-HAR with robust inference, A6 losses, the forecast-evaluation workflow, A8 accuracy versus calibration, C2 an AI answer
- ▣ What you produce: the A answers, one figure and one table per B exercise, the C2 correction table, one interpretation sentence per exercise
- ▣ Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section; background: [Chapter 9](#) and [Franke, Härdle & Hafner \(2019\)](#)
- ▣ Seminar: Antoaneta Amza, PhD candidate, [antoaneta.amza@csie.ase.ro](mailto:antoaneta.amza@csie.ase.ro)

## Exercise Map

| Exercise        | Question                                                                                         | Type                | Worked example     |
|-----------------|--------------------------------------------------------------------------------------------------|---------------------|--------------------|
| A1              | how precise is one day's RV?                                                                     | Solved              | –                  |
| A3              | how often should a noisy price be sampled?                                                       | Solved              | –                  |
| B3              | does SPY show a noise bias at 5 minutes?                                                         | Solved              | A3                 |
| B1              | which SPY days contain a jump?                                                                   | Solved              | –                  |
| A2              | can six returns reveal a jump?                                                                   | Proposed            | B1, A1             |
| A5              | what does HAR impose on an AR(22)?                                                               | Solved              | –                  |
| B5              | how persistent is SPY's log variance?                                                            | Solved              | A5                 |
| A6              | which loss ranks variance forecasts correctly?                                                   | Solved              | –                  |
| A8              | is GARCH less accurate, biased, or both?                                                         | Proposed            | A5, A6             |
| C2              | what is wrong in an AI answer?                                                                   | Proposed            | A3, B5             |
| Extension packs | A2, A3, A5, A6, B1, B3, B5 extended; A4, A7, B2, B4, B6–B9, C1, C3 (optional, after the seminar) | Solved and Proposed | named on each task |

- ☐ **[Solved]:** full solution in the slides and in the notebook, a model to copy; **[Proposed]:** you solve it, the solution is discussed in class
- ☐ Part A: derivations on paper; Part B: computation and inference on SPY and Bitcoin, each ending with an interpretation question; Part C: open questions and AI critique

## Data Used

| Series                        | Content                                                                                  | Sample                                      |
|-------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------|
| SPY, 5-minute bars            | prices of the regular session, UTC time stamps, each bar labelled by its start           | 2020-10-12 – 2026-09-18, 1 489 days         |
| Bitcoin (USD), 5-minute bars  | 24-hour UTC days                                                                         | 659 of 747 days kept                        |
| SPY and Bitcoin, daily prices | adjusted closes (SPY) and closes (Bitcoin), for GARCH                                    | since 2000 (SPY), 2014 (Bitcoin)            |
| SPY daily variances (derived) | intraday RV, overnight return and total variance $V_t$ ( $\%^2$ , $\%$ )                 | one row per SPY day                         |
| One-day forecasts (derived)   | log-HAR, HAR, GARCH(1,1)- $t$ , EWMA, yesterday's RV (column $RW$ ) and the target $V_t$ | from 3 Jan 2022 (SPY), 1 Mar 2025 (Bitcoin) |

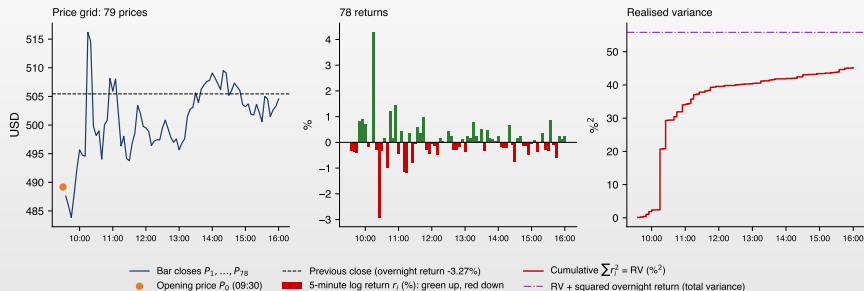
- SPY: UTC converted to New York time; bars 09:30–15:55; missing bars inside a day carry the last price; 78 returns a day, 43 on the 12 half-days
- Bitcoin: days with fewer than 95% of the 288 bars are dropped; short gaps carry the last price
- Units: returns  $r = 100 \Delta \ln P$  in %, variances in  $\%^2$ ; annualise with 252 (SPY) or 365 (Bitcoin) days; all data end on 18 September 2026

## Setup: Run and Verify

```
# notebook section "Data" defines
# spy_intraday, btc_intraday,
# intraday_returns, spy_overnight
P = spy_intraday()          # day x 79 prices
R = intraday_returns(P)     # day x 78 returns, %
print(P.shape)
print(R.notna().sum(axis=1).value_counts())
Pb = btc_intraday()         # day x 289 prices, UTC
print(Pb.shape, Pb.index[0], Pb.index[-1])
```

- Expected output
  - ▶ (1489, 79): SPY days, from 2020-10-12 to 2026-09-18
  - ▶ 1 477 days with 78 returns, 12 with 43
  - ▶ (659, 289): Bitcoin, 2024-09-01 to 2026-09-17
- Column 0 of P is the opening price of the 09:30 bar; columns 1–78 are the closes of the bars
- If your counts differ, stop and check the time zone, the session filter and the cut-off date

## From Bars to Returns: One SPY Day



- 7 April 2025: opening price  $P_0 = 489.19$ , then bar closes  $P_1 = 487.60$ ,  $P_2 = 485.88$ ,  $\dots$ ,  $P_{78} = 504.54$ : 79 prices give 78 returns; differencing the 78 closes alone would give 77
- $r_1 = 100 \ln(P_1/P_0) = -0.33\%$ ,  $r_2 = -0.35\%$ ;  $RV = \sum_i r_i^2 = 45.1\%^2$ ; the overnight return  $-3.27\%$  is outside the bars, total variance  $V = RV + r_{on}^2 = 55.8\%^2$
- Overnight return: the adjusted close-to-close return minus the open-to-close return, so dividends fall in the overnight part

## Notation Used in Every Exercise

- ▣ Day  $t$  has  $M$  intraday returns  $r_{t,i}$  (%);  $RV_t = \sum_i r_{t,i}^2$ ; realised quarticity  $RQ_t = \frac{M}{3} \sum_i r_{t,i}^4$
- ▣ Bipower variation [Barndorff-Nielsen & Shephard \(2004\)](#)  $BV_t = \frac{\pi}{2} \frac{M}{M-1} \sum_{i \geq 2} |r_{t,i}| |r_{t,i-1}|$ ; tripower quarticity  $TQ_t = M \mu_{4/3}^{-3} \frac{M}{M-2} \sum_{i \geq 3} |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$ , a jump-robust estimate of  $IQ_t$ ,  $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$ ,  $Z \sim N(0, 1)$
- ▣  $IV_t$ : integrated variance,  $IQ_t$ : integrated quarticity;  $\omega^2$ : variance of the microstructure noise
- ▣ Total daily variance  $V_t = RV_t + r_{on,t}^2$  for SPY;  $V_t = RV_t$  for Bitcoin (24 hours)
- ▣ Forecast  $F_t$  of  $V_t$ , made at the close of day  $t-1$ ;  $QLIKE(V, F) = V/F - \ln(V/F) - 1$ ;  $MSE = (V - F)^2$ , on variance

## Realised Variance: Volatility Measured after the Fact

- ▣ From inferring to **measuring** daily variance Andersen et al. (2001)
  - ▶ GARCH (Chapter 5) infers today's variance from daily returns through a model
  - ▶ with intraday prices we add up squared 5-minute returns
- ▣ Integrated variance: the variance the day actually delivered
  - ▶  $IV_t = \int_{t-1}^t \sigma_s^2 ds$ ,  $\sigma_s$ : the spot volatility inside the day
- ▣ Realised variance: the sum of squared intraday returns
  - ▶  $RV_t = \sum_{i=1}^M r_{t,i}^2 \rightarrow IV_t$  as  $M \rightarrow \infty$  (continuous price, no noise)
  - ▶  $r_{t,i}$ : the  $i$ -th intraday return of day  $t$ ; SPY:  $M = 78$  five-minute returns a session
- ▣ Measurement error:  $RV_t - IV_t \approx N(0, 2IQ_t/M)$ 
  - ▶  $IQ_t = \int \sigma_s^4 ds$ : integrated quarticity, estimated by  $RQ_t$  (notation slide)
  - ▶ constant volatility: relative standard error  $\sqrt{2/M} = \sqrt{2/78} = 0.16$ , so one day's RV is a noisy measurement
- ▣ Tiny example: returns 0.10, -0.20, 0.10, 0.20 (%)
  - ▶  $RV = 0.01 + 0.04 + 0.01 + 0.04 = 0.10\%$
  - ▶ daily volatility  $\sqrt{0.10} = 0.32\%$ , annualised  $\sqrt{252 \times 0.10} = 5.0\%$
- ▣ Depth: Chapter 9, after the seminar

## Two Things That Distort RV: Noise and Jumps

- **Microstructure noise:** observed log price = efficient price + error
  - ▶  $p_i^{\text{obs}} = p_i^* + \varepsilon_i$ ,  $\text{Var}(\varepsilon_i) = \omega^2$  (bid-ask bounce, price ticks)
  - ▶ each return carries two noise terms:  $\mathbb{E}[\text{RV}^{(n)}] = \text{IV} + 2n\omega^2$  with  $n$  returns
  - ▶ the bias grows with  $n$ ; the variance of the sampling error falls like  $1/n$
  - ▶ hence an optimal sampling frequency [Bandi & Russell \(2008\)](#); in practice 5 minutes is hard to beat [Liu, Patton & Sheppard \(2015\)](#)
  - ▶ signature plot: mean RV against the sampling interval, flat when the noise bias is negligible
- **Jumps:** RV also collects the squared jumps,  $\text{RV}_t \rightarrow \text{IV}_t + \sum J^2$ 
  - ▶ bipower variation [Barndorff-Nielsen & Shephard \(2004\)](#):  $\text{BV}_t = \frac{\pi}{2} \frac{M}{M-1} \sum_{i \geq 2} |r_{t,i}| |r_{t,i-1}| \rightarrow \text{IV}_t$
  - ▶ a jump enters BV only multiplied by a small neighbouring return
  - ▶  $\pi/2 = 1/(\mathbb{E}|Z|)^2$ ,  $Z \sim N(0, 1)$ : rescales products of absolute Normal returns
- Tiny example,  $M = 4$ , returns 0.1, 0.1, 0.1, 1.0 (%)
  - ▶  $\text{RV} = 1.03$ ;  $\text{BV} = \frac{\pi}{2} \cdot \frac{4}{3} (0.01 + 0.01 + 0.10) = 0.25$ ; jump part  $\text{RV} - \text{BV} = 0.78$
- Depth: [Chapter 9](#), after the seminar

## Testing for a Jump: the Ratio Statistic

- $H_0$ : no jump on day  $t$ 
  - ▶ the relative jump measure  $(RV_t - BV_t)/RV_t$  is near 0 under  $H_0$  and large on a jump day
- Ratio statistic [Huang & Tauchen \(2005\)](#)
  - ▶ 
$$z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\theta M^{-1} \max(1, TQ_t/BV_t^2)}}, \theta = \pi^2/4 + \pi - 5 = 0.609$$
  - ▶  $TQ_t$ : tripower quarticity (notation slide), estimates  $IQ_t$  without being inflated by a jump
  - ▶  $TQ_t/BV_t^2$ : how uneven volatility was within the day, floored at 1 (its constant-volatility value)
- Under  $H_0$ ,  $z_t \approx N(0, 1)$ ; one-sided test
  - ▶ reject when  $z_t > 1.645, 2.326$  or  $3.09$  at 5%, 1% or 0.1%
  - ▶  $p_t = 1 - \Phi(z_t)$ ,  $\Phi$ : the standard Normal distribution function
- Many days, many tests (multiple testing, [Chapter 3](#))
  - ▶ with  $N$  days tested, about  $\alpha N$  rejections appear by chance alone: compare the count with  $\alpha N$
- Depth: [Chapter 9](#), after the seminar

## HAR: a Cascade of Three Horizons

- Stylised fact ([Chapter 1](#)): volatility has long memory
  - ▶ the autocorrelation of  $|r_t|$  decays slowly; a GARCH(1,1) forecast reverts at an exponential rate ([Chapter 5](#))
- HAR [Corsi \(2009\)](#): three averages of past RV
  - ▶  $RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(5)} + \beta_m RV_t^{(22)} + \varepsilon_{t+1}$
  - ▶  $RV_t^{(5)}, RV_t^{(22)}$ : means over the last 5 and 22 days;  $\beta_0$ : constant;  $\varepsilon_{t+1}$ : forecast error
  - ▶  $\beta_d, \beta_w, \beta_m$ : weights on yesterday, the last week and the last month
- Intuition: traders with daily, weekly and monthly horizons
  - ▶ three slopes reproduce a slowly decaying memory
  - ▶ log-HAR uses  $y_t = \ln V_t$  in place of  $RV_t$ : positive forecasts, residuals closer to the Normal distribution
- Estimated by OLS with Newey–West (HAC) standard errors ([Chapter 3](#))
  - ▶ the errors are autocorrelated and heteroskedastic
- Tiny example:  $\hat{\beta} = (0.1, 0.4, 0.3, 0.2)$ ,  $RV_t = 2$ ,  $RV_t^{(5)} = 1.5$ ,  $RV_t^{(22)} = 1$ 
  - ▶ forecast  $0.1 + 0.8 + 0.45 + 0.2 = 1.55\%^2$ ; persistence  $\beta_d + \beta_w + \beta_m = 0.9$
- Depth: [Chapter 9](#), after the seminar

## Judging Variance Forecasts: the Tools of Chapter 5

- The true variance is never observed: forecasts are scored against a proxy
  - ▶ here  $V_t = \text{RV}_t + r_{\text{on},t}^2$  for SPY ( $r_{\text{on},t}$ : overnight return), noisy but conditionally unbiased
- Robust losses [Patton \(2011\)](#): rank forecasts as the true variance would
  - ▶ variance MSE  $(V - F)^2$  and QLIKE( $V, F$ ) =  $V/F - \ln(V/F) - 1$ ;  $F$ : the forecast
  - ▶ QLIKE penalises under-prediction more
- Accuracy: Diebold–Mariano [Diebold & Mariano \(1995\)](#)
  - ▶  $t = \bar{d}/\text{se}_{\text{HAC}}(\bar{d})$  on  $d_t = L(V_t, F_t^A) - L(V_t, F_t^B)$  for models A and B
  - ▶  $L$ : QLIKE or MSE;  $\bar{d}$ : mean of  $d_t$ ;  $t < -1.96$ : A significantly better
- Calibration: Mincer–Zarnowitz regression  $V_t = a + bF_t + u_t$ 
  - ▶  $a$ : intercept,  $b$ : slope,  $u_t$ : error; unbiased forecast:  $a = 0$  and  $b = 1$ , tested jointly (Wald)
  - ▶  $b < 1$ : forecasts too extreme
- Out of sample
  - ▶ at each origin  $t - 1$  the model is re-estimated on data up to  $t - 1$  only, then  $F_t$  is compared with  $V_t$
- Depth: [Chapter 9](#), after the seminar

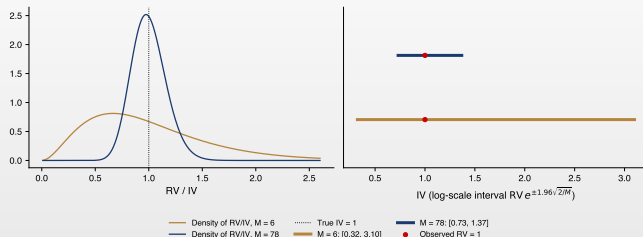
## A1 [Solved]: Precision of Realised Variance — Task

- ▣ **Question:** how precisely does one day of 5-minute returns measure integrated variance?
- ▣ **Inputs:** constant volatility,  $M$  intervals of length  $\Delta = 1/M$ ,  $r_i = \sigma\sqrt{\Delta} Z_i$  with  $Z_i$  i.i.d.  $N(0, 1)$ ,  $IV = \sigma^2$ ; use  $\mathbb{E}Z^2 = 1$ ,  $\text{Var}(Z^2) = 2$ ,  $\mathbb{E}Z^4 = 3$ ;  $\chi_M^2$ : chi-square with  $M$  degrees of freedom (a sum of  $M$  squared i.i.d.  $N(0, 1)$ ); no data file
- ▣ **Tasks**
  1. Show that  $RV = \sigma^2 \chi_M^2 / M$  and derive  $\mathbb{E}[RV]$  and  $\text{Var}(RV)$ .
  2. Show that  $RQ = \frac{M}{3} \sum_i r_i^4$  is unbiased for  $IV = \sigma^4$  and write the estimated standard error of  $RV$ .
  3. Use the delta method for  $\ln RV$  and write the 95% interval for  $IV$  built on the log scale.
  4. Compute the interval factors  $e^{\pm 1.96\sqrt{2/M}}$  for  $M = 78$  and  $M = 6$ , and the relative width for  $M = 78$ .
- ▣ **Report:** the formulas, the two pairs of factors and one sentence on how precise one day's  $RV$  is
- ▣ **Notebook:** section A1

## A1 [Solved]: Solution (1/2) — Derivation

- ▣  $RV = \sum_i \sigma^2 \Delta Z_i^2 = \sigma^2 \frac{1}{M} \sum_i Z_i^2 = \sigma^2 \chi_M^2 / M$ , so  $\mathbb{E}[RV] = \sigma^2$  (unbiased) and  $\text{Var}(RV) = \sigma^4 \cdot 2M/M^2 = 2\sigma^4/M$
- ▣  $\mathbb{E}[r_i^4] = \sigma^4 \Delta^2 \mathbb{E}Z^4 = 3\sigma^4/M^2$ , so  $\mathbb{E}[RQ] = \frac{M}{3} \cdot M \cdot 3\sigma^4/M^2 = \sigma^4$ ; hence  $\widehat{\text{Var}}(RV) = 2RQ/M$  and  $\text{se}(RV) = \sqrt{\frac{2}{3} \sum_i r_i^4}$
- ▣ Delta method with  $g(x) = \ln x$ ,  $g'(x) = 1/x$ :  $\text{se}(\ln RV) \approx \text{se}(RV)/RV = \sqrt{2RQ/M}/RV$
- ▣ 95% interval:  $RV \exp(\pm 1.96 \text{se}(\ln RV))$ , always positive and asymmetric in levels; with constant volatility  $RQ/RV^2 \approx 1$ , so the factors are  $e^{\pm 1.96\sqrt{2/M}}$
- ▣ For volatility  $\sqrt{RV}$ ,  $g'(x) = 1/(2\sqrt{x})$  halves the relative standard error: the interval for volatility is narrower than the one for variance

## A1 [Solved]: Solution (2/2) — Interval Factors and Coverage



- $M = 78$ :  $\sqrt{2/78} = 0.160$ , factors [0.73, 1.37]:  $IV$  lies between 0.73  $RV$  and 1.37  $RV$ , a total relative width of 0.64
- $M = 6$  (30-minute returns): factors [0.32, 3.10]: almost uninformative
- Simulated coverage of the nominal 95% interval with RQ (200 000 days): 94.2% for  $M = 78$ , 86.7% for  $M = 6$ : the 95% is an asymptotic statement
- With stochastic volatility the same studentised limit holds asymptotically, for a continuous price and negligible noise, because the convergence is stable (Chapter 9)
- **Interpretation:** one day of 5-minute data pins  $IV$  down to roughly  $-27\%/+37\%$

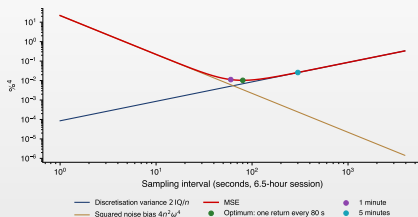
## A3 [Solved]: Noise Bias and the Optimal Frequency — Task

- ▣ **Question:** how often should we sample a price that contains noise?
- ▣ **Inputs:**  $p_i^{\text{obs}} = p_i^* + \varepsilon_i$ ,  $\varepsilon_i$  i.i.d.  $(0, \omega^2)$  independent of  $p^*$ ; zero drift, constant volatility;  $IV = 1\%^2$ ,  $IQ = 1$ ,  $\omega = 0.01\%$ , so  $\omega^2 = 0.0001\%^2$ ; session of 23 400 seconds
- ▣ **Tasks**
  1. Write the observed return, expand  $\mathbb{E}[r_i^2]$  and derive  $\mathbb{E}[RV^{(n)}] = IV + 2n\omega^2$  for  $n$  returns.
  2. Derive  $\text{cov}(r_i, r_{i-1}) = -\omega^2$ .
  3. From  $\text{MSE}(n) \approx 2IQ/n + 4n^2\omega^4$  [Bandi & Russell \(2008\)](#), derive the MSE-optimal  $n^*$  and its sampling interval.
  4. Fill a table for 5 minutes, 1 minute, 1 second and  $n^*$ :  $n$ , bias  $2n\omega^2$ ,  $\mathbb{E}[RV]$ , MSE.
- ▣ **Report:** the derivations, the table and which interval you would use in practice, with the reason
- ▣ **Notebook:** section A3

## A3 [Solved]: Solution (1/2) — Derivation

- $r_i = p_i^{\text{obs}} - p_{i-1}^{\text{obs}} = r_i^* + \varepsilon_i - \varepsilon_{i-1}$
- $\mathbb{E}[r_i^2] = \mathbb{E}[r_i^{*2}] + \mathbb{E}[\varepsilon_i^2] + \mathbb{E}[\varepsilon_{i-1}^2] = \mathbb{E}[r_i^{*2}] + 2\omega^2$  (cross terms have mean 0 by independence); summing over  $n$  returns:  $\mathbb{E}[\text{RV}^{(n)}] = \text{IV} + 2n\omega^2$
- Adjacent returns share only  $\varepsilon_{i-1}$ , with opposite signs:  $\text{cov}(r_i, r_{i-1}) = \text{cov}(-\varepsilon_{i-1}, \varepsilon_{i-1}) = -\omega^2$
- $\frac{d}{dn} \text{MSE} = -2\text{IQ}/n^2 + 8n\omega^4 = 0$  gives  $n^3 = \text{IQ}/(4\omega^4)$ ,  $n^* = (\text{IQ}/(4\omega^4))^{1/3}$ ; the second derivative is positive: a minimum
- Numbers:  $n^* = (1/(4 \times 10^{-8}))^{1/3} = 292$ , one return every  $23\,400/292 = 80$  seconds

## A3 [Solved]: Solution (2/2) — Table and Chart



| Interval           | $n$    | $2n\omega^2$ | $\mathbb{E}[\text{RV}]$ | MSE     |
|--------------------|--------|--------------|-------------------------|---------|
| 5 minutes          | 78     | 0.0156       | 1.0156                  | 0.0259  |
| 1 minute           | 390    | 0.0780       | 1.0780                  | 0.0112  |
| 1 second           | 23 400 | 4.6800       | 5.6800                  | 21.9025 |
| $n^*$ , every 80 s | 292    | 0.0585       | 1.0585                  | 0.0103  |

- $\text{MSE} = 2/n + 4n^2 \times 10^{-8}$  in %<sup>4</sup>; at one second RV is almost six times IV
- In this model 1 minute is close to the optimum; 5 minutes has 2.5 times the minimal MSE
- **Interpretation:** the usual 5-minute choice rests on arguments outside this model
  - ▶  $\omega$  is uncertain; real noise is not i.i.d.
  - ▶ 5-minute RV is hard to beat empirically [Liu, Patton & Sheppard \(2015\)](#)

## B3 [Solved]: Which Sampling Frequency? — Task

▣ **Worked example to follow:** A3 [Solved]

▣ **Questions**

- ▶ does SPY show a measurable noise bias at 5 minutes?
- ▶ which grid gives the most useful daily measure?

▣ **Data:** the SPY price matrix  $P$  of the setup (1 489 days); grid of  $k \times 5$  minutes,  $k = 1, 3, 6, 13$  (5, 15, 30, 65 minutes): prices at columns  $o, o + k, o + 2k, \dots$  plus the open and the close, for offsets  $o = 0, \dots, k - 1$  (helper `sparse_points`)

▣ **Tasks**

1. Compute daily RV on one grid ( $o = 0$ ) and the average over the  $k$  shifted grids, and report  $\sqrt{252 \overline{RV}}$  for each interval.
2. For each interval compute the ratio of mean one-grid RV to mean 5-minute RV and a 95% moving-block bootstrap interval [Künsch \(1989\)](#): resample 20-day blocks of the paired daily series, 2 000 times, and recompute the ratio of means.
3. Compute the correlation of today's log RV (one grid and averaged grids) with tomorrow's 5-minute log RV.

▣ **Report:** one table (volatility, ratio [CI], correlation per interval) and one chart

▣ **Interpretation**

- ▶ is there a measurable noise bias at 5 minutes?
- ▶ why a block bootstrap and not an i.i.d. one?
- ▶ which grid is most useful for forecasting?

▣ **Notebook:** section B3

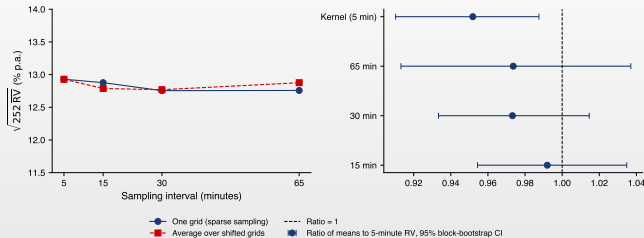
## B3 [Solved]: Solution (1/2) — Procedure and Table

- Procedure: for each day take the grid prices, compute  $RV^{(o)} = \sum (100 \Delta \ln P)^2$  on each offset  $o$ , average over  $o$ ; for the bootstrap, draw block starts at random, glue 20-day blocks of the paired series  $(RV_t^{(k)}, RV_t^{(s)})$  to the original length, take  $\sum RV^{(k)} / \sum RV^{(s)}$ , and read the 2.5% and 97.5% percentiles

| Interval | Vol. (one grid) | Vol. (avg. grids) | Ratio [95% CI]       | Corr. next day | Corr. (avg.) |
|----------|-----------------|-------------------|----------------------|----------------|--------------|
| 5 min    | 12.93           | 12.93             | 1.000 [1.000, 1.000] | 0.73           | 0.73         |
| 15 min   | 12.88           | 12.79             | 0.992 [0.954, 1.035] | 0.71           | 0.72         |
| 30 min   | 12.75           | 12.77             | 0.973 [0.933, 1.015] | 0.70           | 0.72         |
| 65 min   | 12.76           | 12.88             | 0.974 [0.913, 1.037] | 0.64           | 0.70         |

- Block bootstrap: daily RVs are strongly autocorrelated; i.i.d. resampling would destroy this dependence and make the interval too narrow

## B3 [Solved]: Solution (2/2) — Chart and Answer



- All intervals contain 1: the coarse grids detect no difference from 5-minute RV; the signature is flat
- The predictive correlation falls from 0.73 (5 minutes) to 0.64 (65 minutes, one grid): sparse RV is noisier; averaging the shifted grids recovers part of it (0.70)
- **Interpretation:** 5-minute RV is the most useful measure here; in the kernel comparison (extension pack) the realised kernel averages 0.952 of the 5-minute RV (95% CI [0.910, 0.987]): a small but significant difference

## B1 [Solved]: Jump Days in SPY — Task

### □ Questions

- ▶ on which SPY days does the ratio test reject?
- ▶ how many rejections would chance alone produce?

- **Data:** the SPY return matrix  $R$  (1 489 days;  $M_t = 78$ , or 43 on half-days); ratio test [Huang & Tauchen \(2005\)](#), formulas on the notation slide; helper `jump_test`

### □ Tasks

1. For every day compute  $RV$ ,  $BV$ ,  $TQ$ ,  $z_t$  and the one-sided  $p_t = 1 - \Phi(z_t)$ ; work 27 August 2021 step by step.
2. Count the days that reject at 5%, 1% and 0.1% and compare each count with  $\alpha N$ .
3. On the 0.1% days find the interval with the largest  $|r_{t,i}|$  and plot the histogram of its end time.
4. Compute the jump share  $\sum_t J_t / \sum_t RV_t$ , with  $J_t = RV_t - BV_t$  on 0.1% days and 0 otherwise.

- **Report:** the worked day, a table of counts against  $\alpha N$ , the histogram and the jump share

### □ Interpretation

- ▶ are there more rejections than the level alone would produce?
- ▶ what could explain them?

- Notebook: section B1

## B1 [Solved]: Solution (1/3) — One Day Step by Step

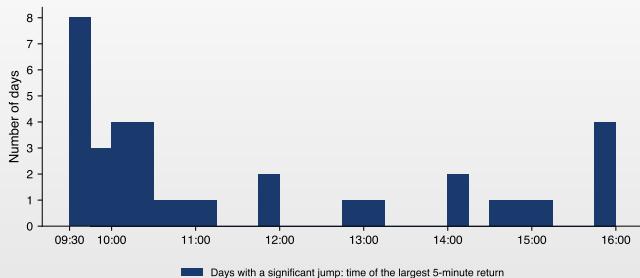
- ▣ 27 August 2021,  $M = 78$ :  $RV = 0.1354$ ,  $BV = 0.0687$ ,  $TQ = 0.00539$  ( $\%^2$ ,  $\%^2$ ,  $\%^4$ )
- ▣  $(RV - BV)/RV = 0.493$ ;  $TQ/BV^2 = 1.143 > 1$ , so the floor does not bind
- ▣  $z = 0.493/\sqrt{0.609 \times 1.143/78} = 5.22$ ,  $p = 9.0 \times 10^{-8}$ : reject at 0.1%
- ▣ The largest return, +0.24% in the interval ending 10:05, is 44% of RV: 10:00–10:05 is when the Federal Reserve chair's Jackson Hole speech was released
- ▣ The same four steps, vectorised over days, are `jump_test(R)`

## B1 [Solved]: Solution (2/3) — Counts and Jump Share

| Level | Rejections | $\alpha N$ |
|-------|------------|------------|
| 5%    | 235        | 74.5       |
| 1%    | 110        | 14.9       |
| 0.1%  | 35         | 1.5        |

- $z_t$  has mean 0.50 and standard deviation 1.14 instead of 0 and 1: far more rejections than the level implies
- Jump share of total RV with 0.1% days: 2.8%
- Explanations: real jumps, and the intraday U-shape, which biases BV down (extension pack: FDR control and the periodicity correction)

## B1 [Solved]: Solution (3/3) — When Do the Largest Returns Occur?



- Time of the largest  $|r|$  on flagged days: it peaks after the open; attributing a flag to news needs separate evidence

## A2 [Proposed]: Six Returns and One Jump — Task

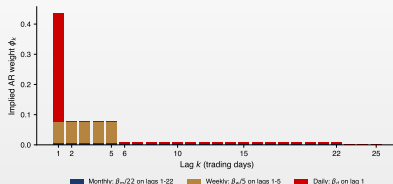
- ▣ **Worked example to follow:** B1 [Solved] (one day step by step) and A1
- ▣ **Question:** can six 5-minute returns reveal a jump?
- ▣ **Inputs:** the returns in the table (%); the last one is a news jump;  
 $TQ = 0.0097$  (supplied);  $\theta = \pi^2/4 + \pi - 5 = 0.609$ ; critical value at 0.1%: 3.09
- ▣ **Tasks**
  1. Complete the table and compute RV and  $BV = \frac{\pi}{2} \frac{M}{M-1} \sum_{i \geq 2} |r_i r_{i-1}|$  with  $M = 6$ .
  2. Compute the jump part  $RV - BV$  and its share of RV.
  3. Compute  $TQ/BV^2$ , apply the floor  $\max(1, TQ/BV^2)$  and compute 
$$z = \frac{(RV - BV)/RV}{\sqrt{\theta M - 1} \max(1, TQ/BV^2)}.$$
  4. Decide at 0.1% and compare  $z$  with its largest possible value  $\sqrt{M/\theta}$ .
- ▣ **Report:** the completed table, RV, BV,  $z$ , the decision and one sentence of interpretation

| $i$ | $r_i$ | $r_i^2$ | $ r_i r_{i-1} $ |
|-----|-------|---------|-----------------|
| 1   | 0.08  | 0.0064  | –               |
| 2   | –0.15 |         |                 |
| 3   | 0.03  |         |                 |
| 4   | 0.10  |         |                 |
| 5   | –0.04 |         |                 |
| 6   | –0.90 |         |                 |

## A5 [Solved]: HAR as a Restricted AR(22) — Task

- **Question:** which restrictions does HAR impose on an AR(22) for log variance?
- **Inputs:** log-HAR [Corsi \(2009\)](#)  $y_{t+1} = \beta_0 + \beta_d y_t + \beta_w \bar{y}_t^{(5)} + \beta_m \bar{y}_t^{(22)} + \varepsilon_{t+1}$ ,  $y_t = \ln V_t$ ,  $\bar{y}_t^{(5)}$  the mean of  $y_t, \dots, y_{t-4}$ ,  $\bar{y}_t^{(22)}$  the mean of  $y_t, \dots, y_{t-21}$ ; rounded SPY estimates  $\hat{\beta} = (-0.08, 0.36, 0.34, 0.17)$
- **Tasks**
  1. Expand the two means and write the implied AR coefficients  $\phi_1, \phi_2 = \dots = \phi_5$  and  $\phi_6 = \dots = \phi_{22}$ .
  2. With  $\theta = (\phi_0, \phi_1, \dots, \phi_{22})$ , write two rows of the matrix  $R$  in  $R\theta = 0$  and count the restrictions.
  3. Compute  $\phi_1, \phi_{2..5}, \phi_{6..22}$  and the persistence  $\sum_k \phi_k$  for  $\hat{\beta}$ , and draw the 22 weights.
- **Report:** the three weight formulas, two rows of  $R$ , the numbers and what the persistence implies for shocks
- Notebook: section A5

## A5 [Solved]: Solution



- Lag 1 is in all three components:  $\phi_1 = \beta_d + \beta_w/5 + \beta_m/22$ ; lags 2–5 in the weekly and monthly ones:  $\phi_k = \beta_w/5 + \beta_m/22$ ; lags 6–22 only in the monthly one:  $\phi_k = \beta_m/22$
- Rows of  $R$  (column 0: intercept):  $(0, 0, 1, -1, 0, \dots, 0)$  for  $\phi_2 = \phi_3$ ;  $(0, \dots, 0, 1, -1, 0, \dots)$  at columns 6, 7 for  $\phi_6 = \phi_7$
- Count: 3 equalities among  $\phi_2, \dots, \phi_5$  plus 16 among  $\phi_6, \dots, \phi_{22}$ : 19 rows,  $R$  is  $19 \times 23$  (22 lags minus 3 free slopes)
- $\phi_1 = 0.36 + 0.34/5 + 0.17/22 = 0.436$ ;  $\phi_{2..5} = 0.076$ ;  $\phi_{6..22} = 0.008$ ;  $\sum_k \phi_k = \beta_d + \beta_w + \beta_m = 0.87$
- **Interpretation:** persistence  $0.87 < 1$ : shocks to log variance decay slowly but die out; the data test of the 19 restrictions is in the extension pack

## B5 [Solved]: log-HAR for SPY with Robust Inference — Task

- ▣ **Worked example to follow:** A5 [Solved]
- ▣ **Questions**
  - ▶ how persistent is SPY's log total variance?
  - ▶ do robust standard errors change any conclusion?
- ▣ **Data:** SPY total daily variance from the setup,  $V_t = RV_t + r_{\text{on},t}^2$ , 1 489 days; regressors dated  $t - 1$ :  $\ln V_{t-1}$  and the means of  $\ln V$  over days  $t - 5..t - 1$  and  $t - 22..t - 1$  (mean of logs); the first 22 days are lost
- ▣ **Tasks**
  1. Build the design matrix and check your row for 8 April 2025 against the solution.
  2. Estimate log-HAR by OLS; report the usual and the Newey–West (HAC) standard errors with 7 lags, rule  $\lfloor 4(T/100)^{2/9} \rfloor$ .
  3. Test  $\beta_w = \beta_m$  with a Wald test using the Newey–West covariance.
  4. Run Ljung–Box tests with 22 lags on  $\ln V_t$  and on the residuals, plot both ACFs, and compare  $R^2$  with an AR(1).
- ▣ **Report:** the coefficient table with both standard errors, the tests with  $p$ -values and the chart
- ▣ **Interpretation**
  - ▶ which component matters most?
  - ▶ do robust errors change a conclusion?
  - ▶ has HAR captured the persistence?
- ▣ **Notebook:** section B5

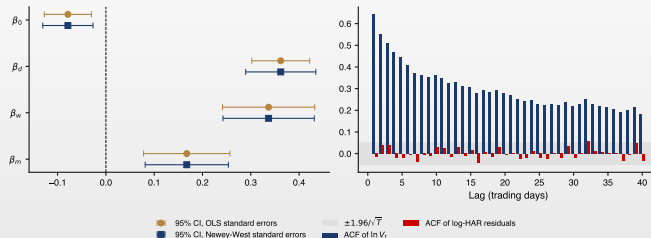
## B5 [Solved]: Solution (1/2) — Design Row and Estimates

- Row for 8 April 2025:  $y_t = \ln V_t = 2.976$ ; regressors  $\ln V_{t-1} = 4.022$  (7 April), weekly mean 2.068, monthly mean 0.612
- Fitted value  $\hat{y}_t = 2.177$ : the model expects a partial return towards the monthly level
- Usable observations:  $1\,489 - 22 = 1\,467$ ;  $R^2 = 0.470$  against 0.413 for an AR(1) ( $\hat{\phi} = 0.643$ ); persistence 0.866

|           | $\hat{\beta}$ | SE OLS | SE NW | $t_{\text{NW}}$ |
|-----------|---------------|--------|-------|-----------------|
| $\beta_o$ | -0.079        | 0.025  | 0.027 | -3.0            |
| $\beta_d$ | 0.362         | 0.031  | 0.037 | 9.8             |
| $\beta_w$ | 0.337         | 0.049  | 0.048 | 7.0             |
| $\beta_m$ | 0.167         | 0.046  | 0.044 | 3.8             |

- Newey–West errors are up to 20% larger than the OLS ones; all slopes stay significant

## B5 [Solved]: Solution (2/2) — Tests and Chart

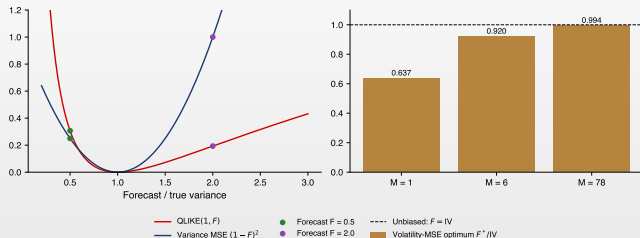


- Wald  $\beta_w = \beta_m$ : 4.17,  $p = 0.041$ : rejected at 5%, the weekly weight is larger
- Ljung–Box(22):  $\ln V_t$  4641, residuals 17.5 ( $p = 0.73$ ): three coefficients absorb the persistence; first-order residual autocorrelation  $-0.014$
- Interpretation:** the weekly and daily components dominate; robust errors do not change a conclusion; HAR leaves no significant autocorrelation

## A6 [Solved]: Which Loss Ranks Variance Forecasts? — Task

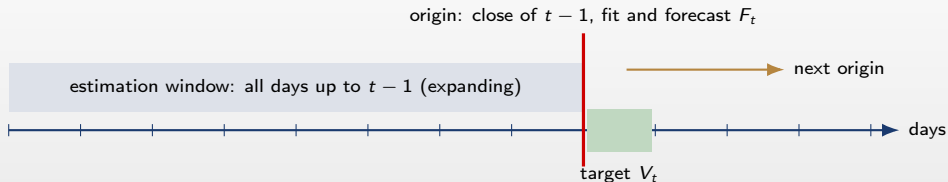
- **Question:** which loss ranks variance forecasts correctly when the target is a noisy proxy?
- **Inputs:** proxy  $V_t$  conditionally unbiased:  $\mathbb{E}[V_t | \mathcal{F}_{t-1}] = \mathbb{E}[IV_t | \mathcal{F}_{t-1}]$  [Patton \(2011\)](#);  
QLIKE( $V, F$ ) =  $V/F - \ln(V/F) - 1$ ; variance MSE  $(V - F)^2$
- **Tasks**
  1. Minimise  $\mathbb{E}[\text{QLIKE}(V_t, F) | \mathcal{F}_{t-1}]$  over  $F$  and show that the minimiser is  $\mathbb{E}[IV_t | \mathcal{F}_{t-1}]$ .
  2. For two forecasts  $F_A, F_B$  show that  $\Delta L = V(1/F_A - 1/F_B) + \ln(F_A/F_B)$  is linear in  $V$ , so its conditional mean is the same with  $V_t$  or with  $IV_t$ : the ranking is preserved.
  3. With true variance 1, compute QLIKE and variance MSE for  $F = 0.5$  and  $F = 2$  and compare the two rankings.
- **Report:** the first-order condition, the ranking argument, four loss values and which loss you would use
- **Notebook:** section A6

## A6 [Solved]: Solution



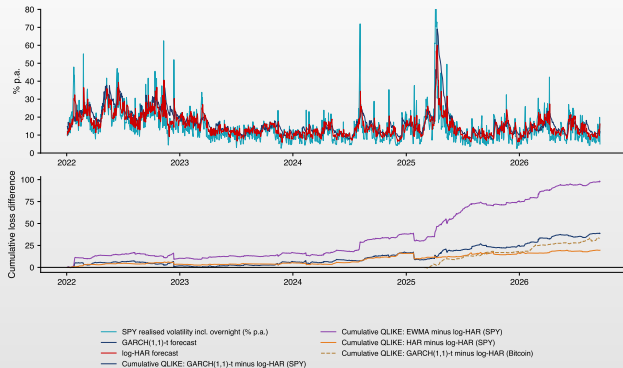
- $\partial_F \mathbb{E}[V/F - \ln V + \ln F - 1] = -\mathbb{E}V/F^2 + 1/F = 0$  gives  $F = \mathbb{E}[V_t | \mathcal{F}_{t-1}] = \mathbb{E}[IV_t | \mathcal{F}_{t-1}]$ ; second derivative  $1/F^2 > 0$
- $\text{QLIKE}(V, F_A) - \text{QLIKE}(V, F_B) = V(1/F_A - 1/F_B) + \ln(F_A/F_B)$ : linear in  $V$ , and  $F_A, F_B$  are known at  $t - 1$ , so replacing  $V_t$  by the conditionally unbiased  $IV_t$  leaves  $\mathbb{E}[\Delta L | \mathcal{F}_{t-1}]$  unchanged
- QLIKE:  $F = 0.5$  gives  $2 - \ln 2 - 1 = 0.307$ ,  $F = 2$  gives  $0.5 + \ln 2 - 1 = 0.193$ ; variance MSE: 0.25 and 1, the opposite ranking
- **Interpretation:** both losses are robust, but QLIKE penalises under-prediction more, which matters for risk; MSE on volatility is not robust (extension pack)

## Forecast Evaluation: Origin, Target, Window



- At each origin: re-estimate on all data up to  $t - 1$ , forecast  $F_t$ , then observe  $V_t$ ; no value from day  $t$  enters the forecast
- SPY: 1 180 origins from 3 January 2022 to 18 September 2026; losses compared on common dates only
- One-day forecasts of log-HAR, HAR, GARCH(1,1)- $t$ , EWMA and yesterday's RV (column RW), and the target  $V_t$

## One-Day Forecasts and Cumulative Losses



- Mean QLIKE: log-HAR 0.310, GARCH(1,1)-t 0.343; a rising cumulative line means log-HAR has the lower loss
- $d_t = \text{QLIKE}(\text{log-HAR}) - \text{QLIKE}(\text{GARCH})$  is the input of A8: is the gap more than chance?

## A8 [Proposed]: Accuracy versus Calibration — Task

- ▣ **Worked example to follow:** A5 (Wald statistic) and A6 [Solved]
- ▣ **Question:** is GARCH less accurate than log-HAR, biased, or both?
- ▣ **Inputs:** SPY one-day forecasts, January 2022 – September 2026;  $d_t = \text{QLIKE}(\log\text{-HAR}) - \text{QLIKE}(\text{GARCH})$  has mean  $\bar{d} = -0.0327$ , Newey–West SE 0.0142; MZ regression for GARCH,  $V_t = a + bF_t + u_t$ :  $\hat{a} = 0.2418$  (SE 0.2310),  $\hat{b} = 0.7046$  (SE 0.2382),  $\text{corr}(\hat{a}, \hat{b}) = -0.9750$  (keep four decimals)
- ▣ **Tasks**
  1. Accuracy,  $H_0$ : equal expected loss. Compute the Diebold–Mariano  $t$ , its two-sided  $p$ -value, and decide at 5%.
  2. Calibration,  $H_0$ :  $a = 0$ ,  $b = 1$ . Compute the separate  $t$ -statistics for  $a = 0$  and  $b = 1$ .
  3. Build the  $2 \times 2$  covariance matrix, compute the Wald statistic  $W = \delta^\top \hat{V}^{-1} \delta$  (the distance of the estimates from  $(0, 1)$ , weighted by their covariance  $\hat{V}$ ) with  $\delta = (\hat{a}, \hat{b} - 1)$  and its  $\chi^2_2$   $p$ -value.
  4. Draw the 95% confidence ellipse, mark  $(0, 1)$  and explain when separate and joint tests disagree.
- ▣ **Report:** three statistics, three decisions and one sentence separating accuracy from calibration

## C2 [Proposed]: Critique an AI Answer — Task

- ▣ **Worked example to follow:** A3 and B5 [Solved]; a code-reading exercise, no tick data are needed
- ▣ **Setting:** a student gives an AI assistant a pandas Series `px` of clean, time-sorted SPY trade prices in New York time, 09:30–16:00, several days, in USD (so returns are decimal, not %)
- ▣ **Tasks**
  1. The answer on the next slide contains three conceptual errors; everything else is correct under this setting. Find them.
  2. For each error give the evidence: a derivation, a check on the output, or a number from the lecture or this seminar.
  3. Give the correction: formula, code line and the corrected number where one is asked.
  4. Log each error you actually found in the format of `AI_ERRORS.md`: request, answer, error, how found, correction.
- ▣ **Report:** a table with one row per error (what, evidence, correction)
- ▣ Hints: A3 on  $\mathbb{E}[RV^{(n)}]$  with noise; what happens between 16:00 and 09:30; plug in  $RV_t = 1.0 \times 10^{-4}$  ( $1\%^2$ )

## C2 [Proposed]: Critique an AI Answer — the Answer

- ▣ **Request:** “Compute the daily realised variance, the annualised volatility and the HAR regressors from px.”
- ▣ **Answer, text**
  - ▶  $RV_t = \sum_i r_{t,i}^2$  with  $r_{t,i}$  the intraday log returns of day  $t$ ; use every tick: more returns always make  $RV_t$  more precise
  - ▶  $RV_t$  is SPY's daily variance: compare it directly with squared close-to-close returns and with the GARCH variance
  - ▶ HAR: regress  $RV_{t+1}$  on  $RV_t$  and on its 5-day and 22-day averages [Corsi \(2009\)](#)

```
import numpy as np, pandas as pd
lp = np.log(px)
r = lp.groupby(lp.index.date).diff().dropna()
RV = (r**2).groupby(r.index.date).sum()
vol_ann = np.sqrt(RV) * 252
H = pd.DataFrame({'d': RV,
                  'w': RV.rolling(5).mean(),
                  'm': RV.rolling(22).mean(),
                  'y': RV.shift(-1)}).dropna()
```

## Take-Home Choice

### Take-home: choose one

- ▣ B2, jumps in Bitcoin (model: B1), or B6, forecasting the next week (model: B5, A8)
- ▣ Write it as a Quantlet folder (code, `Metainfo.txt`, the chart): the format the team project uses
- ▣ AI use is allowed; note the errors you actually catch, in the format of `AI_ERRORS.md`
- ▣ Extension packs are optional

## Extension Packs: What Each One Needs

| Pack             | Exercises                                          | Needs                                            |
|------------------|----------------------------------------------------|--------------------------------------------------|
| 1. Noise         | A3 extended [Solved], A4, B3 extended [Solved]     | A3, B3                                           |
| 2. Jumps         | A2 extended, B1 extended [Solved], B2              | A2, B1                                           |
| 3. Distributions | B4                                                 | B3 (block bootstrap)                             |
| 4. HAR           | A5 extended [Solved], B5 extended [Solved], A7, B7 | A5, B5, B1                                       |
| 5. Forecasting   | A6 extended [Solved], B6, B6 extended, B9          | A6, A8, the evaluation workflow                  |
| 6. Roughness     | B8                                                 | B3, <a href="#">Chapter 9</a> (rough volatility) |
| 7. Projects      | C1, C3                                             | B5, A6, B9, the lecture case study               |

- Each exercise names its notebook section; within a pack, work in order
- C1 and C3 can seed the team project

## A3 Extended [Solved]: Estimating the Noise Variance — Task

### ▣ Questions

- ▶ how can  $\omega^2$  be estimated from the data?
- ▶ in which limit is each estimator consistent?

▣ **Inputs:** the model of A3,  $n$  returns per day

### ▣ Tasks

1. Show that  $\mathbb{E}[\text{RV}^{\text{all}}/(2n)] = \omega^2 + \text{IV}/(2n)$  and state its limit as  $n \rightarrow \infty$ .
2. Propose a second estimator from the first-order autocovariance and state when it is consistent.

▣ Notebook: section A3

## A3 Extended [Solved]: Solution

- $\mathbb{E}[\text{RV}^{\text{all}} \mid \sigma] = \text{IV} + 2n\omega^2$ , so  $\mathbb{E}[\text{RV}^{\text{all}}/(2n)] = \omega^2 + \text{IV}/(2n)$ ; a law of large numbers for the noise terms gives  $\hat{\omega}^2 = \text{RV}^{\text{all}}/(2n) \xrightarrow{P} \omega^2$  as  $n \rightarrow \infty$  [Bandi & Russell \(2008\)](#)
- $-\widehat{\text{cov}}(r_i, r_{i-1})$  estimates  $\omega^2$  (A3), consistent as the number of returns grows, provided the noise is i.i.d. and independent of the price
- In finite samples the first estimator is contaminated by  $\text{IV}/(2n)$ ; A4 compares it with the oracle value

## A4 [Proposed]: Two-Scale RV — Task

▣ **Worked example to follow:** A3 [Solved]

▣ **Inputs:** a synthetic illiquid stock with a stipulated 390-minute session:  $IV = IQ = 1$ ,  $\omega = 0.03\%$ ; one day: 1-minute returns ( $n = 390$ ) give  $RV^{\text{all}} = 1.70$ ;  $K = 5$  shifted 5-minute grids give  $RV^{\text{avg}} = 1.10$

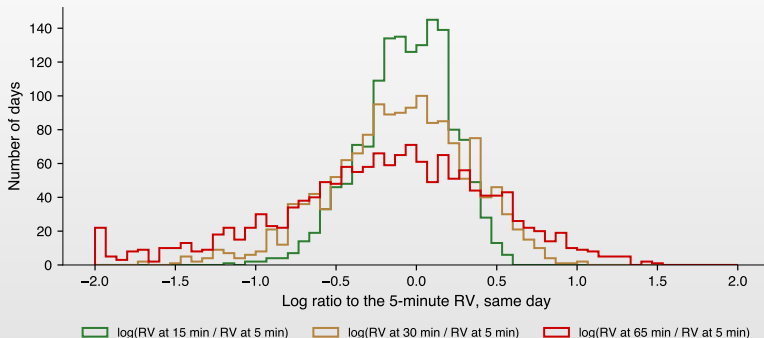


1. Compute the noise bias  $2n\omega^2$  at 5 and 1 minutes, and  $n^*$  of A3 with its interval.
2. With  $\bar{n} = (n - K + 1)/K$ , compute  $TSRV = RV^{\text{avg}} - \frac{\bar{n}}{n}RV^{\text{all}}$ , with and without the factor  $1/(1 - \bar{n}/n)$ .
3. Estimate  $\omega^2$  twice: the oracle  $(RV^{\text{all}} - IV)/(2n)$  and the feasible  $RV^{\text{all}}/(2n)$  of A3 extended.
4. Show that TSRV removes the noise bias to first order. Optional: with  $\text{Var}(\text{error}) \approx aK/n + bn/K^2$ , find the  $K$  that balances the terms and the rate of TSRV.

## B3 Extended [Solved]: A Realised Kernel on 5-Minute Bars — Task

- ▣ **Question:** does a noise-robust estimator change SPY's 5-minute RV?
- ▣ **Data:** SPY returns  $R$  and prices  $P$ ; helper `realized_kernel` [Barndorff-Nielsen et al. \(2008\)](#): Parzen weights, bandwidth  $H = \lceil 3.5134 \xi^{4/5} n^{3/5} \rceil$  per day ( $\lceil \cdot \rceil$ : round up), noise-to-signal ratio  $\xi^2 = \hat{\omega}^2 / \widehat{IV}$ , the course grid endpoints (open and close)
- ▣ **Tasks**
  1. Compute the kernel for every day and the ratio of its mean to mean RV, with the block bootstrap of B3.
  2. Plot the distribution of  $\ln(RV^{(k)} / RV^{(5)})$  for 15, 30 and 65 minutes.
- ▣ **Notebook:** section B3

## B3 Extended [Solved]: Solution



- Kernel / RV = 0.952, CI [0.910, 0.987] (median bandwidth  $H = 7$ ): the kernel adds back small negative autocorrelations ( $\hat{\rho}_1 = -0.008$ ); significant but economically minor
- Coarser grids scatter around the 5-minute value: sparse RV adds noise without removing a detectable bias

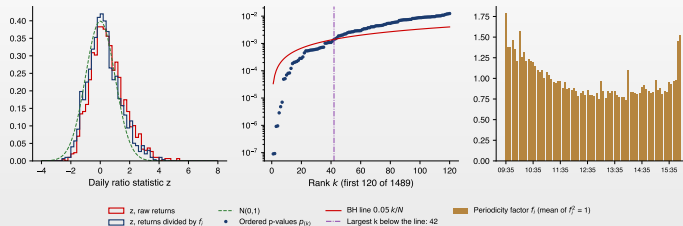
## A2 Extended [Proposed]: Why BV Ignores a Jump — Task

- ▣ **Worked example to follow:** A1 [Solved]
- ▣ **Inputs:** one jump of size  $\kappa$  in an **interior** interval  $j$  ( $1 < j < M$ ):  $r_j = \kappa + r_j^c$ ; the other returns are diffusive,  $O_p(\Delta^{1/2})$
- ▣ **Tasks**
  1. Show that to leading order the jump enters BV only through  $|r_{j-1}| |\kappa| + |\kappa| |r_{j+1}|$ , and that these products are  $O_p(\Delta^{1/2})$ .
  2. Conclude that  $BV \rightarrow IV$  while  $RV \rightarrow IV + \kappa^2$ ; how does a jump in the last interval (as in A2) differ?

## B1 Extended [Solved]: Multiplicity and Intraday Seasonality — Task

- ▣ **Question:** how many of B1's rejections survive a false-discovery-rate control and a correction for the intraday U-shape?
- ▣ **Data:** the  $N = 1\,489$  daily  $p$ -values of B1; helpers `bh_count`, `periodicity_sd`
- ▣ **Tasks**
  1. Benjamini–Hochberg [Benjamini & Hochberg \(1995\)](#): sort the  $p$ -values and reject the  $k$  smallest,  $k$  the largest index with  $p_{(k)} \leq kq/N$ ; apply  $q = 5\%$  and  $1\%$ , and Bonferroni ( $p < 0.05/N$ ).
  2. Periodicity factor [Boudt, Croux & Laurent \(2011\)](#): scale returns by  $\sqrt{\text{BV}_t/M_t}$ , take the root mean square of each 5-minute slot, normalise so that the mean of  $f_i^2$  is 1; divide  $r_{t,i}$  by  $f_i$  and repeat B1(2) and step 1.
- ▣ **Interpretation:** BH controls the FDR for valid  $p$ -values that are independent or positively dependent [Benjamini & Yekutieli \(2001\)](#); which count would you use to build a jump regressor?

## B1 Extended [Solved]: Solution



- BH at 5%: the largest  $k$  with  $p_{(k)} \leq k \times 0.05/1489$  is  $k = 42$  ( $p_{(k)} = 0.00140$  against 0.00141): 42 jump days; at 1%: 9; Bonferroni: 7
- After dividing by  $f_i$  ( $f = 1.78$  at the open, 0.73 at the minimum): 167 rejections at 5%, 20 at 0.1%, 17 under FDR 5%
- **Interpretation:** build the jump regressor from FDR-controlled, periodicity-corrected days

## B2 [Proposed]: Jumps in Bitcoin — Task

- **Worked example to follow:** B1 [Solved]
- **Data:** Bitcoin price matrix  $P_b$ , 24-hour UTC days with at least 95% of the 288 bars, 659 days of 747; a day is labelled by its UTC date
- **Tasks**
  1. Apply the ratio test of B1 at 0.1% and build a daily table: date,  $M_t$ , RV, BV,  $z$ , weekend flag.
  2. For weekdays and weekends report the number of days, the number of rejections and their rate, and the mean BV/RV.
  3. Compute the jump share of total variation and a one-sided binomial test of the overall rate against 0.001.
- **Interpretation:** the binomial test assumes independent, correctly sized daily tests
  - ▶ what does a rejection establish?
  - ▶ what does it not establish?
  - ▶ which group rejects more often?
  - ▶ why might that group reject more often?

## B4 [Proposed]: Returns Standardised by Realised Volatility — Task

- ▣ **Worked example to follow:** B3 [Solved] (block bootstrap)
- ▣ **Data:** SPY open-to-close returns (1 489 days) and Bitcoin 24-hour returns (659 days), each aligned with the same-day RV Andersen et al. (2001); one row: 7 April 2025,  $r_t = 3.09\%$ ,  $RV_t = 45.1$ ,  $z_t = r_t / \sqrt{RV_t} = 0.46$
- ▣ **Tasks**
  1. Compute the Pearson kurtosis  $\mathbb{E}(x - \bar{x})^4 / s^4$  of  $u_t = r_t / \text{sd}(r)$  and of  $z_t$ , with 95% intervals from a paired moving-block bootstrap (20-day blocks, 2 000 resamples).
  2. Plot histograms and Normal QQ plots of  $u_t$  and  $z_t$ ; run Jarque–Bera on  $z_t$ ; compute the share of  $|z_t| > 3$  (Normal: 0.27%).
- ▣ **Interpretation:** where do the heavy tails come from? Optional benchmark: for i.i.d. Gaussian intraday returns  $z_t$  has kurtosis  $3M/(M + 2)$

## A5 Extended [Solved]: The Wald Test of the HAR Restrictions — Task

▣ **Worked example to follow:** A5 [Solved]

▣ **Tasks**

1. Write the Wald statistic for  $R\theta = 0$  with a Newey–West covariance  $\hat{V}_{NW}$  and give its limiting distribution under HAR.
2. Explain why a HAC covariance is needed for the log-variance regression.

## A5 Extended [Solved]: Solution

- ▣  $W = (R\hat{\theta})^\top (R\hat{V}_{\text{NW}}R^\top)^{-1}R\hat{\theta} \rightarrow \chi_{19}^2$  under the 19 HAR restrictions
- ▣ The errors of the log-variance regression are heteroskedastic and may be autocorrelated: the usual OLS covariance would misstate  $\text{Var}(R\hat{\theta})$
- ▣ The data test is B5 extended, step 1

## B5 Extended [Solved]: The AR(22) Test and HARQ — Task

- ▣ **Worked example to follow:** A5 extended, B5 [Solved]
- ▣ **Data:** as in B5; realised quarticity  $RQ_t$  from the intraday returns (it does not cover the overnight return)
- ▣ **Tasks**
  1. Estimate the unrestricted AR(22) of  $\ln V_t$  and test the 19 HAR restrictions with a Newey–West Wald test.
  2. Estimate HARQ in levels [Bollerslev, Patton & Quaadvlieg \(2016\)](#),  
$$V_{t+1} = \beta_0 + (\beta_d + \beta_Q \sqrt{RQ_t})V_t + \beta_w V_t^{(w)} + \beta_m V_t^{(m)} + \varepsilon$$
, on intraday RV and on total variance.
  3. Forecast from 3 January 2022 on an expanding window, replacing forecasts that fall outside the  $[\min, \max]$  range of the in-sample values by the sample mean [Bollerslev, Patton & Quaadvlieg \(2016\)](#), and run DM against HAR in levels under QLIKE.
- ▣ **Notebook:** section B5

## B5 Extended [Solved]: Solution

- ▣ (1)  $W = 18.1$ ,  $p = 0.51$  ( $\chi^2_{19}$ );  $R^2$  0.477 (AR(22)) vs 0.470 (HAR): the restrictions are not rejected; HAR saves 19 parameters
- ▣ (2)  $\hat{\beta}_Q = -0.0088$  (SE 0.0042) on intraday RV, significant at 5%;  $-0.0022$  (SE 0.0024) on total variance, not significant
- ▣ (3) QLIKE 0.344 (HARQ) vs 0.326 (HAR), DM  $t = 1.23$ ,  $p = 0.22$ : no significant difference
- ▣ **Interpretation:** the sign predicted by A7 appears, but with 5-minute RV the measurement error is small and the overnight return, not covered by RQ, dominates the noise in total variance

## A7 [Proposed]: Attenuation and the Case for HARQ — Task

- ▣ **Worked example to follow:** A5 [Solved], B5 extended
- ▣ **Inputs:**  $RV_t = V_t^* + u_t$  (latent signal plus error),  $V_t^*$  the integrated variance;  $\mathbb{E}[u_t | \sigma] = 0$ ,  $\text{Var}(u_t | \sigma) = 2 \text{IQ}_t / M$  (asymptotic, A1); errors serially uncorrelated and uncorrelated with  $V^*$  and  $\eta$ ;  $V_{t+1}^* = \beta_0 + \beta V_t^* + \eta_{t+1}$ ,  $\beta > 0$
- ▣ **Tasks**
  1. Using  $\text{plim } \hat{\beta} = \text{cov}(RV_{t+1}, RV_t) / \text{Var}(RV_t)$ , derive the attenuation of the OLS slope.
  2. Illustration (signal variance  $\sigma_{V^*}^2$  assumed not to depend on  $\text{IQ}_t$ ): derive the best linear weight on  $RV_t$  given  $\text{IQ}_t$  and show that it falls with  $\text{IQ}_t$ .
  3. Write an instrumental-variables estimator with instrument  $RV_{t-1}$  and state its exclusion and relevance conditions.

## B7 [Proposed]: Do Jumps Help? The HAR-CJ Model — Task

- ▣ **Worked example to follow:** B5 and B1 [Solved]
- ▣ **Data:** SPY as in B5; replication baseline: B1's 0.1% jump days, without corrections;  $J_t = RV_t - BV_t$  on those days, 0 otherwise;  $C_t = RV_t - J_t + r_{\text{on},t}^2$  (all in %<sup>2</sup>) [Andersen, Bollerslev & Diebold \(2007\)](#)
- ▣ **Tasks**
  1. Regress  $\ln V_{t+1}$  on  $\ln C_t$ ,  $\ln C_t^{(5)}$ ,  $\ln C_t^{(22)}$ ,  $\ln(1 + J_t)$ ,  $\ln(1 + J_t^{(5)})$ ,  $\ln(1 + J_t^{(22)})$ , where (5), (22) are arithmetic means logged afterwards; report all coefficients with Newey–West SEs and test the three jump coefficients jointly.
  2. Forecast from 3 January 2022 on an expanding window and compare QLIKE with log-HAR on common dates by DM.
  3. Sensitivity: repeat with the FDR-controlled, periodicity-corrected jump days of B1 extended.
- ▣ **Interpretation:** HAR-CJ averages levels before logging, log-HAR averages logs: the two are not nested; why can a significant jump coefficient give no forecasting gain?

## A6 Extended [Solved]: Why MSE on Volatility Is Not Robust — Task

- ▣ **Worked example to follow:** A6 [Solved]
- ▣ **Inputs:** constant IV and  $RV = IV \chi_M^2 / M$ ; supplied moment:  $\mathbb{E}\sqrt{\chi_M^2} = \sqrt{2} \Gamma(\frac{M+1}{2}) / \Gamma(\frac{M}{2})$ ,  $\Gamma$ : the gamma function
- ▣ **Tasks**
  1. Minimise  $\mathbb{E}(\sqrt{RV} - \sqrt{F})^2$  (MSE on volatility) over  $F$  and compare the minimiser with  $\mathbb{E}RV$  by Jensen's inequality.
  2. Compute  $F^*/IV$  for  $M = 1$  (squared daily return), 6 and 78.

## A6 Extended [Solved]: Solution

- $\mathbb{E}(\sqrt{RV} - \sqrt{F})^2 = \mathbb{E}RV - 2\sqrt{F}\mathbb{E}\sqrt{RV} + F$ , minimal at  $\sqrt{F} = \mathbb{E}\sqrt{RV}$ ; by Jensen  $(\mathbb{E}\sqrt{RV})^2 < \mathbb{E}RV$ : the optimal forecast is biased down
- $F^*/IV = \frac{2}{M} \left( \Gamma(\frac{M+1}{2}) / \Gamma(\frac{M}{2}) \right)^2 = 0.637$  ( $M = 1$ ), 0.920 ( $M = 6$ ), 0.994 ( $M = 78$ ); plotted in A6
- **Interpretation:** against squared daily returns, MSE on volatility rewards a forecast 36% too low; with 5-minute RV the distortion almost disappears

## B6 [Proposed]: Forecasting the Next Week — Task

- ▣ **Worked example to follow:** B5 [Solved], A8, the evaluation workflow
- ▣ **Data:** SPY total variance  $V_t$  (B5) and SPY adjusted closes since 2000 for GARCH; target  $Y_t = V_t + \dots + V_{t+4}$ , forecast at the close of  $t - 1$ ; 1 176 origins from 3 January 2022
- ▣ **Tasks**
  1. Direct log-HAR: regress  $\ln Y$  on the B5 regressors, re-estimated at every origin using only targets fully observed by  $t - 1$  (the last four are not); forecast  $\exp(\hat{y} + \hat{s}^2/2)$  [Duan \(1983\)](#).
  2. GARCH(1,1)- $t$  re-estimated every 21 days: with  $\sigma_t^2$  the one-step variance,  $\omega$ ,  $\alpha$ ,  $\beta$  the GARCH constant, reaction and persistence coefficients and  $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$  the long-run variance, forecast  $\sum_{h=0}^4 [\bar{\sigma}^2 + (\alpha + \beta)^h (\sigma_t^2 - \bar{\sigma}^2)]$ .
  3. Compute QLIKE and MSE, DM on QLIKE with HAC errors, both MZ coefficients with the joint test of  $(a, b) = (0, 1)$ , and plot the cumulative QLIKE difference.
- ▣ **Interpretation**
  - ▶ does the one-day advantage of log-HAR survive at a weekly horizon?
  - ▶ what do overlapping targets do to the DM test?

## B6 Extended [Proposed]: A Conditional Giacomini–White Test — Task

▣ **Worked example to follow:** B6

▣ **Tasks**

1. Re-estimate both methods on fixed-length rolling windows (log-HAR: last 250 targets; GARCH: last 1 000 returns), as the theory of [Giacomini & White \(2006\)](#) requires.
2. With instruments  $Z_t = (1, d_{t-5})$  and the loss difference  $d_{t+5}$  realised five days later, compute  $n \bar{g}^\top \hat{\Omega}^{-1} \bar{g}$  with  $g = Z_t d_{t+5}$  ( $n$ : number of origins,  $\bar{g}$ : the mean of  $g$ ,  $\hat{\Omega}$ : its long-run covariance); under the null  $g$  is uncorrelated beyond lag 4, so  $\hat{\Omega}$  sums the unweighted autocovariances of lags 0–4.

## B9 [Proposed]: Which Models Survive? The Model Confidence Set — Task

- ▣ **Worked example to follow:** B5 extended, B6
- ▣ **Data:** SPY one-day forecasts (log-HAR, HAR, GARCH(1,1)-t, EWMA, yesterday's RV) plus HARQ (B5 extended) and SHAR [Patton & Sheppard \(2015\)](#):  $\beta_d V_t$  replaced by  $\beta^+ RS_t^+ + \beta^- RS_t^-$ ,  $RS_t^\pm = \sum_i r_{t,i}^2 1\{r_{t,i} \gtrless 0\}$ , overnight return counted as one more return; 1 180 common dates
- ▣ **Tasks**
  1. Build the date  $\times$  model QLIKE matrix on common dates and check that every forecast is positive.
  2. Run the MCS [Hansen, Lunde & Nason \(2011\)](#) with the range statistic, 20-day blocks, 10 000 resamples, seed 42: `MCS(L, size=0.10, reps=10000, block_size=20, method='R', seed=42)`.
  3. Report the elimination order and every model's MCS  $p$ -value; compare the sets at 90% (size 0.10) and 75% (size 0.25).
- ▣ **Interpretation:** why is the MCS preferable to 21 pairwise DM tests?

## B8 [Proposed]: How Rough Is Volatility? — Task

- ▣ **Worked example to follow:** B3 [Solved] (block bootstrap), [Chapter 9](#)
- ▣ **Data:** SPY daily series of four measures: total variance, intraday RV, BV, 30-minute subsampled RV;  $x_t = \frac{1}{2} \ln(\text{measure}_t)$ ; lags  $\Delta = 1, \dots, 30$  trading days [Gatheral, Jaisson & Rosenbaum \(2018\)](#)
- ▣ **Tasks**
  1. Compute  $m(q, \Delta)$ , the mean of  $|x_{t+\Delta} - x_t|^q$  over all valid pairs, for  $q \in \{0.5, 1, 1.5, 2, 3\}$ ; check  $m(2, 1)$  for total variance.
  2. For each  $q$  regress  $\ln m(q, \Delta)$  on  $\ln \Delta$  with an intercept: slope  $\zeta_q$ ; then  $H = \text{slope of } \zeta_q \text{ on } q \text{ through the origin.}$
  3. 95% CI: moving-block bootstrap, 60-day blocks, 300 resamples, increments computed only within each block.
  4. Simulate a Gaussian random walk (increments  $N(0, 0.1^2)$ , seed 42), remove the line joining its endpoints (endpoint-detrended, still  $H = 0.5$ ), estimate  $H$ , add i.i.d.  $N(0, 0.25^2)$  noise, estimate again.
- ▣ **Interpretation**
  - ▶ is  $H$  below 0.5 for every measure?
  - ▶ what does measurement error do?
- ▣ One path is an illustration, not a size or power study

## C1 [Proposed]: Is Bitcoin's Volatility More Predictable Than the S&P 500's? — Task

- **Data:** SPY and Bitcoin one-day forecasts, common calendar window 1 March 2025 – 18 September 2026 (567 calendar days: SPY 389 trading days, Bitcoin 422 days)
- **Tasks**
  1. Define three measures and compute them per asset: the calibration  $R^2$  of  $\ln V_t$  on  $\ln F_t$ ; the QLIKE gain  $1 - \overline{\text{QLIKE}}_{\text{log-HAR}} / \overline{\text{QLIKE}}_{\text{RV yest.}}$ ; the same gain over GARCH.
  2. Build a 95% CI for the  $R^2$  difference (Bitcoin minus SPY): resample the same 28-calendar-day blocks for both assets (2 000 times), keeping each asset's own missing days.
  3. Add a dummy for a weekend target day to the Bitcoin log-HAR, forecast again and run DM under QLIKE with and without it.
- **Report:** the table, the CI, the DM result and a two-sentence answer; project extension: Ethereum or a BVB stock, and the entropy measure of [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)

## C3 [Proposed]: Does Machine Learning Beat HAR for SPY? — Project Brief (1/4)

- ▣ Context: Christensen, Siggaard & Veliyev (2023) (Chapter 9, case study)
  - ▶ random forests and neural networks beat HAR once predictors beyond past RV enter
  - ▶ the gain grows with the horizon
  - ▶ you test this on SPY, with the sample split and the model settings of the paper
- ▣ (a) Build the target series (Section 4 of the paper).
  - ▶ daily open-to-close RV of SPY from the 5-minute bars of the regular session (78 returns a day, as in B1)
  - ▶ three targets:  $RV_{t+1}$ , the average RV over the next 5 days and over the next 22 days
- ▣ (b) Build two information sets dated  $t$ .
  - ▶  $\mathcal{M}_{\text{HAR}}$ : the daily, weekly and monthly RV
  - ▶  $\mathcal{M}_{\text{ALL}}$  adds the VIX, the squared daily log return of the Hang Seng and the 1-week SPY momentum
  - ▶ and the first differences of log dollar volume and of the 3-month T-bill rate (FRED, DTB3)
  - ▶ and the economic policy uncertainty index (FRED, USEPUINXD) and the Aruoba–Diebold–Scotti index (Philadelphia Fed)

### C3 [Proposed]: Does Machine Learning Beat HAR for SPY? — Project Brief (2/4)

- (b, continued) Date each predictor by its release, not its reference date.
  - ▶ use only values published by the close of day  $t$
  - ▶ the Aruoba–Diebold–Scotti index is revised: use the Philadelphia Fed's real-time vintages
  - ▶ disclose any series for which vintages are unavailable
  - ▶ state which predictors of the paper (Table 1) cannot be built for an index fund, and why: a data limitation, not a reason to change the method
- (c) Split the sample as in Appendix A.1 of the paper.
  - ▶ the first 1000 usable days for training, the next 200 for validation, the rest for the test
  - ▶ a row belongs to a subset only if its whole target window lies inside it: drop the last  $h - 1$  rows before each boundary
  - ▶ report the usable sample size per horizon
  - ▶ standardise every predictor with the training mean and variance only

## C3 [Proposed]: Does Machine Learning Beat HAR for SPY? — Project Brief (3/4)

- ▣ (d) Estimate the linear models.
  - ▶ HAR, log-HAR and OLS on all predictors, on a rolling window over training and validation merged
  - ▶ log-HAR forecast:  $\exp(\hat{f} + \hat{\sigma}^2/2)$ ,  $\hat{f}$ : fitted log RV,  $\hat{\sigma}^2$ : residual variance
  - ▶ elastic net on a rolling training window, without merging;  $\lambda$  and the mixing weight chosen on the validation set (paper, Section 2)
  - ▶ HAR keeps its three RV regressors under both information sets
- ▣ (d, continued) Estimate the machine-learning models (paper, Table A.6).
  - ▶ random forest: 500 trees on the merged window,  $\lfloor J/3 \rfloor$  candidate predictors per split ( $J$ : number of predictors), minimum node size 5
  - ▶  $\text{NN}_2^{10}$ : layers of 4 and 2 neurons, leaky ReLU, Adam with learning rate 0.001, dropout 0.8
  - ▶ early stopping with patience 100; the average of the best 10 of 100 seeds; trained once on a fixed window
  - ▶ replace negative forecasts with the smallest in-sample RV

## C3 [Proposed]: Does Machine Learning Beat HAR for SPY? — Project Brief (4/4)

- (e) Report the test-set comparison, as in B9.
  - ▶ a table of MSE and QLIKE relative to HAR, for the three horizons and both information sets
  - ▶ one-sided DM  $p$ -values against HAR and the 90% MCS
  - ▶ overlapping 5- and 22-day targets: HAC errors with a bandwidth above  $h - 1$  (Bartlett weights down-weight lag  $h - 1$ ) and a sensitivity check over bandwidths
- Interpretation
  - ▶ does the ranking of Tables 2, 3 and 7 of the paper survive on SPY?
  - ▶ does the ML advantage grow with the horizon?
  - ▶ does the ML advantage survive QLIKE?
  - ▶ with a test set of this length, what can a non-significant DM test tell you?
- Worked example to follow: B5, A6.
- Project extension
  - ▶ repeat on Bitcoin's 24-hour RV
  - ▶ compare across models the importance of RV and VIX from accumulated local effects (Figure 7 of the paper)

## References (1/2)

- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X. (2007). [Roughing it up: including jump components in the measurement, modeling, and forecasting of return volatility](#). *Review of Economics and Statistics*, 89(4), 701–720.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2001). [The distribution of realized exchange rate volatility](#). *Journal of the American Statistical Association*, 96(453), 42–55.
- ▣ Bandi, F.M., Russell, J.R. (2008). [Microstructure noise, realized variance, and optimal sampling](#). *Review of Economic Studies*, 75(2), 339–369.
- ▣ Barndorff-Nielsen, O.E., Hansen, P.R., Lunde, A., Shephard, N. (2008). [Designing realized kernels to measure the ex post variation of equity prices in the presence of noise](#). *Econometrica*, 76(6), 1481–1536.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2004). [Power and bipower variation with stochastic volatility and jumps](#). *Journal of Financial Econometrics*, 2(1), 1–37.
- ▣ Benjamini, Y., Hochberg, Y. (1995). [Controlling the false discovery rate: a practical and powerful approach to multiple testing](#). *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300.
- ▣ Benjamini, Y., Yekutieli, D. (2001). [The control of the false discovery rate in multiple testing under dependency](#). *The Annals of Statistics*, 29(4), 1165–1188.
- ▣ Bollerslev, T., Patton, A.J., Quaedvlieg, R. (2016). [Exploiting the errors: a simple approach for improved volatility forecasting](#). *Journal of Econometrics*, 192(1), 1–18.
- ▣ Boudt, K., Croux, C., Laurent, S. (2011). [Robust estimation of intraweek periodicity in volatility and jump detection](#). *Journal of Empirical Finance*, 18(2), 353–367.
- ▣ Christensen, K., Siggaard, M., Veliyev, B. (2023). [A machine learning approach to volatility forecasting](#). *Journal of Financial Econometrics*, 21(5), 1680–1727.
- ▣ Corsi, F. (2009). [A simple approximate long-memory model of realized volatility](#). *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy](#). *Journal of Business & Economic Statistics*, 13(3), 253–263.

## References (2/2)

- ▣ Duan, N. (1983). Smearing estimate: a nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383), 605–610.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- ▣ Huang, X., Tauchen, G. (2005). The relative contribution of jumps to total price variance. *Journal of Financial Econometrics*, 3(4), 456–499.
- ▣ Künsch, H.R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3), 1217–1241.
- ▣ Liu, L.Y., Patton, A.J., Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1), 293–311.
- ▣ Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- ▣ Patton, A.J., Sheppard, K. (2015). Good volatility, bad volatility: signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3), 683–697.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- ▣ Zhang, L., Mykland, P.A., Aït-Sahalia, Y. (2005). A tale of two time scales: determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411.